CAP182 | DATA SCIENCE PROJECT

From registration to first deposit

An end-to-end, lifecycle-based explanation of the STADIOEquities account activation analysis

PROJECT QUESTION

Will an account make its first deposit within 90 days of registration?

4,906	9 of 12	0.943	98.6%
unique accounts	tables selected	test ROC-AUC, RF	outreach-list precision

EXECUTIVE TAKEAWAY

Saved test results show useful retrospective ranking performance. Random Forest has the highest test F1 (0.857) and recall (0.943), while Logistic Regression has the highest accuracy (0.870). The ensemble-based lowest-score 30% segment contains 218 of 221 non-activators (98.6% precision), covering 51.7% of test-set non-activators. These are predictive associations, not evidence that outreach causes deposits or that the model is ready for live use.

Prepared for project interpretation and reproducibility

Report generated 2026-10-25 | Bernette Mogwera Lesejane

The data science lifecycle

A CRISP-DM-style map: evaluation findings can send the work back for revision.

1 Business understanding

Define 90-day activation and the need to prioritise support.

2 Data understanding

Audit 12 tables, quality, target balance, relevance and leakage.

3 Data preparation

Clean, aggregate, split, cap, impute, scale and encode.

4 Modelling

Train Logistic Regression, Random Forest, XGBoost and ensemble.

5 Evaluation

Test performance, ranking, sensitivity and repeated CV.

6 Deployment / action

Pilot a controlled outreach policy and monitor outcomes.

Iteration matters

The feature-importance and sensitivity results raise a timing question about KYC fields. Confirm when each predictor exists relative to the intended scoring moment, revise the feature set if necessary, then repeat evaluation. This is a feedback loop between data understanding, preparation and evaluation - not a one-way checklist.

Scope

This report interprets the repository notebook and saved CSV/figure outputs. It distinguishes measured predictive performance from business impact, which still requires a prospective intervention test.

1-2. Business and data understanding

Define the outcome, establish the population and decide which source tables are fit for purpose.

Target definition

Binary account-level outcome: activated = 1 if the first deposit occurs within 90 days of registration; otherwise activated = 0. A low predicted activation probability is treated as greater risk of stalling.

Population and class balance

Table 01 has 4,920 rows. Removing 14 exact duplicates leaves 4,906 unique accounts: 2,093 activated (42.7%) and 2,813 not activated (57.3%). The majority-to-minority ratio is 1.34:1, a moderate imbalance.

Stratified split

Subset	Accounts	Activated	Rate
Training	3,434	1,465	42.7%
Validation	736	314	42.7%
Unseen test	736	314	42.7%

Table relevance

Nine tables are used fully or partially: registration/onboarding; initial session; demographics/identity; communications/support; pre-deposit interactions; in-app messaging; aggregated marketing; KYC/compliance; and device/technical. Excluded: support tickets (no timestamp to establish pre-deposit timing and duplicate Table 04 counts), feedback (barrier label is leakage-prone with incomplete, non-random response), and post-deposit activity (only exists for activated accounts). Gender is excluded as a protected attribute.

3. Data preparation and code walkthrough

Create one defensible modelling row per account and keep fitting decisions inside the training data.

Paths and ingestion

`BASE` and `DATA_DIR` locate the project and source folder. `Path.glob("table*.csv")` discovers the tables. `read_csv(..., keep_default_na=False, na_values=[""])` preserves the literal category "None" while empty cells become missing. The inventory records rows, columns, unique IDs and exact duplicates.

Selection, cleaning and aggregation

A relevance table records inclusion decisions. Category variants are canonicalised across case/spacing. Numeric values are coerced and checked. Marketing rows are grouped by account into exposure count, clicks, average spend, channel count and click rate.

Leakage controls

Deposit date and days-to-deposit are outcomes or outcome-derived, so they are excluded. Pre-deposit durations censored by deposit, constant-zero balance, campaign conversion/ROI and raw campaign identifiers/timestamps are also removed. The code checks that forbidden columns are absent.

Split and transform

Stratified 70/15/15 split: 3,434 train, 736 validation, 736 test. IDs are checked for overlap. Numeric 99th-percentile caps, median imputation, scaling and categorical imputation/one-hot encoding are fit on training data, then applied to validation/test. The model matrix contains 169 features.

Why it matters

If preprocessing or oversampling learns from the test set, evaluation becomes optimistic. Fitting caps and the transformer on training only, then holding the test set aside, helps estimate performance on unseen accounts.

4. Modelling: what the code does

Three model families provide a performance comparison; averaged probabilities provide a combined ranking.

Feature transformation

The target is separated from predictors. `ColumnTransformer` applies median imputation and standardisation to numeric columns and fills missing categories with "Not stated" before one-hot encoding. `handle_unknown="ignore"` supports categories that appear only after fitting.

Models and training

SMOTE with k=5 is applied to training data only; validation/test retain observed prevalence. Random Forest uses 100 trees, depth 15 and balanced class weights. XGBoost uses 100 rounds, learning rate 0.1, depth 6 and subsampling. Logistic Regression is the simpler baseline. Seed 42 supports repeatability.

Ensemble and decision rule

The ensemble averages the RF, XGBoost and Logistic Regression activation probabilities. A 0.5 threshold creates class labels for metrics. Separately, the outreach analysis sorts by ensemble probability and selects the lowest 30%; that is a capacity-based ranking rule, not the classification threshold.

CODE IN PLAIN LANGUAGE

```
X_train_bal, y_train_bal = SMOTE(k_neighbors=5, random_state=42).fit_resample(X_train, y_train)
rf_model.fit(X_train_bal, y_train_bal)
p_ensemble = (p_rf + p_xgb + p_lr) / 3
```

SMOTE is correctly applied after splitting; in cross-validation it is placed inside the pipeline so each fold resamples only its training portion.

5. Evaluation: unseen-test results

The held-out test set has 736 accounts at the same 42.7% activation rate as train and validation.

Model	Accuracy	Prec.*	Recall*	F1	ROC-AUC	Brier
Random Forest	0.865	0.785	0.943	0.857	0.943	0.093
XGBoost	0.852	0.795	0.879	0.835	0.940	0.098
Logistic Reg.	0.870	0.810	0.908	0.856	0.937	0.095
Ensemble	0.861	0.794	0.911	0.849	0.942	0.093

Metric meaning

Precision and recall treat activation (class 1) as positive. Precision: among predicted activators, how many did activate? Recall: among actual activators, how many were identified? F1 balances the two. ROC-AUC measures ranking across thresholds; lower Brier means better probability accuracy.

Results interpretation

Random Forest leads test F1 (0.857), recall (0.943) and ROC-AUC (0.943). Logistic Regression leads accuracy (0.870) and precision (0.810). The gaps are modest. The best model depends on the decision objective, not one metric alone.

Generalisation warning

Tree-model training scores are near-perfect compared with test: XGBoost F1 0.998 train vs 0.835 test; ensemble 0.979 vs 0.849. This gap signals overfitting or a difficult generalisation step. Use held-out performance, not training fit, to assess likely future behaviour.

How well does the model rank accounts?

ROC-AUC compares ranking performance over all possible thresholds.

Test-set evidence

Test ROC-AUC is 0.943 for Random Forest, 0.942 for the ensemble, 0.940 for XGBoost and 0.937 for Logistic Regression. All are materially above the 0.5 random-ranking reference; the curves are close, so differences are not large.

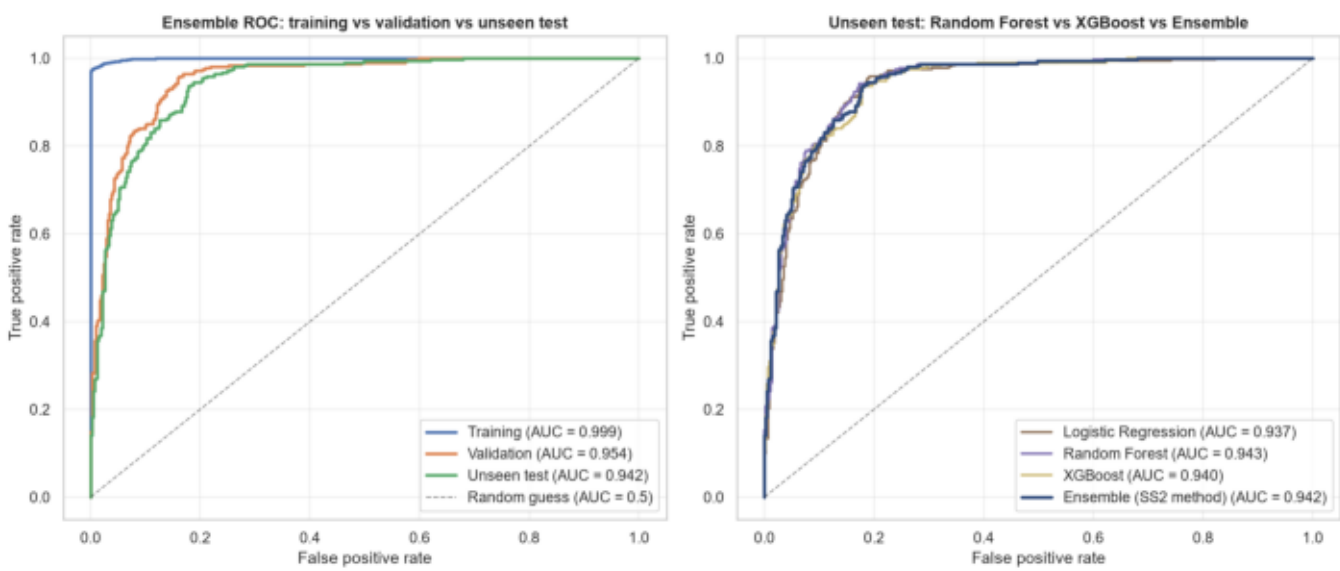


Figure: ROC curves from saved project outputs. ROC-AUC does not specify outreach workload, positive predictive value at a chosen capacity, or causal impact.

Business context

A strong ranking can help order accounts for limited support capacity. To make a decision, select an operating capacity or cost-sensitive threshold, inspect the corresponding errors, and test interventions. AUC alone is not a launch criterion.

Operational interpretation: an outreach list

The project evaluates a concrete capacity scenario using the 30% lowest ensemble activation scores.

Measure on unseen test	Observed
Accounts selected	221
Non-activators in test	422
Non-activators reached	218
Coverage of non-activators	51.7%
Selected-list precision for stalling	98.6%
Random-list baseline precision	57.3%

Meaning

The selected 221 accounts contain 218 non-activators, while reaching 218 of 422 total non-activators. Thus the list is highly concentrated with stalling accounts but still misses about 204 of them. The random-targeting baseline is the class prevalence, not a causal comparison.

Confusion matrix translated

At threshold 0.5 for the ensemble, saved counts imply 286 correctly predicted activators, 28 actual activators flagged as non-activators, 348 correctly identified non-activators and 74 non-activators predicted to activate. In business terms, some likely stallers are missed and some would receive unnecessary outreach.

Recommended test

Treat 30% as a capacity assumption. Run a controlled pilot comparing outreach with business-as-usual among eligible high-risk accounts. Measure incremental 90-day activation, cost per incremental activation, opt-outs and complaints. The retrospective precision does not prove that outreach changes behaviour.

What the model uses - and what to verify

Feature importance is predictive contribution under fitted models, not evidence of causality or intervention value.

Leading signals

The top ensemble-average signals include KYC verification recorded (~0.215), days to KYC verification (~0.099), onboarding completed (~0.091), time to onboarding completion (~0.055), verified KYC status (~0.042), and early session, email, deposit-page and push-engagement measures.

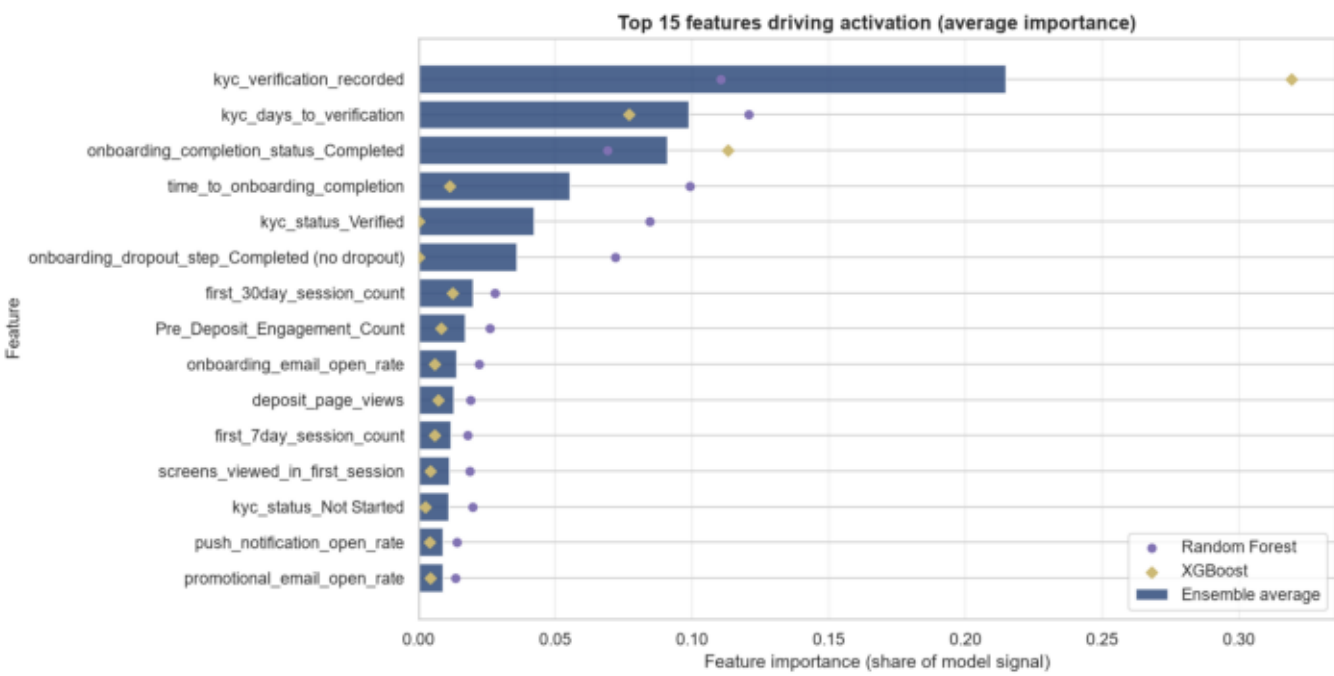


Figure: top 15 features by average model importance from saved outputs.

Point-in-time risk

KYC timing/status are snapshot-sensitive and prominent. Confirm every feature existed by the scoring time and before the outcome window. A feature recorded after activation can act as a proxy for the label even if it is not explicitly target-derived. The notebook removes direct deposit leakage, but a statistical screen cannot prove temporal validity.

Sensitivity result

Removing timing-sensitive snapshots: F1 rises from 0.849 to 0.858 while AUC changes from 0.942 to 0.940. Removing additional pre-deposit counts: F1 is 0.852 and AUC 0.929. Similar performance is reassuring but requires source-system timestamp validation.

Robustness, next steps and handover

Use the statistical comparison as supporting evidence, then complete prospective and governance checks.

Model	Mean CV F1	Reported 95% interval
Logistic Regression	0.848	0.844-0.853
Random Forest	0.852	0.849-0.856
XGBoost	0.848	0.844-0.852

Statistical comparison

Repeated stratified CV uses 5 folds x 10 repeats. Random Forest mean F1 is 0.852, Logistic Regression 0.848 and XGBoost 0.848; reported intervals overlap. Paired t-tests report RF differences versus Logistic Regression (p=0.011) and XGBoost (p=0.0046), while reported Mann-Whitney comparisons are not significant (p >= 0.13). Repeated folds are dependent, so treat these p-values cautiously; evidence does not establish a decisive winner.

Lifecycle actions before deployment

1) Define the exact scoring time and audit feature timestamps. 2) Validate on later registration cohorts and, where possible, a different population. 3) Pilot outreach with a randomised holdout to measure incremental deposits. 4) Monitor calibration, errors, base rate, missingness and drift. 5) Review permitted use, privacy, fairness and human oversight.

Reproduction and artifacts

Run the notebook top-to-bottom with the 12 `data/table*.csv` files and documented requirements. `artifacts/` contains fitted models, preprocessing and split/prediction files; `results/` contains audit, metric and statistical CSVs; `results/figures/` contains saved visuals. This report uses saved repository outputs; re-run and refresh them when code/data changes.

Conclusion

The work establishes a promising retrospective signal and a specific candidate outreach ranking. It does not yet establish a causal uplift or prospective deployment readiness. A validated point-in-time feature set and controlled pilot are the next lifecycle stages.